

EVORA, Portugal

WMO: 085570

Lat: 38.567N

Lon: 7.900W

Elev: 1053

StdP: 14.15

Time Zone: 0.00 (EUW)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.9	39.3	24.2	19.1	44.2	27.4	22.2	46.7	27.0	51.8	23.2	49.9	9.8	320	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	96.2	68.0	92.6	66.6	89.3	65.9	69.4	90.4	68.3	87.8	67.1	84.9	7.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	93.3	70.4	62.8	89.0	70.5	61.6	85.2	69.7	34.0	90.8	33.0	87.9	32.1	85.4	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	20.1	18.3	33.9	100.5	3.1	2.6	31.7	102.4	29.9	103.9	28.1	105.4	25.9	107.2
			30.8	71.5	3.2	1.3	28.4	72.5	26.6	73.3	24.8	74.0	22.4	75.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	48.5	50.5	54.6	56.0	61.4	67.9	74.0	74.0	72.5	63.3	55.5	50.9	(6)
	DBStd	10.62	3.89	4.84	5.07	5.56	6.26	6.27	6.73	5.61	6.24	5.74	5.00	4.00	(7)
	HDD50	189	72	47	12	9	1	0	0	0	0	0	12	36	(8)
	HDD65	2542	512	407	325	275	148	37	3	1	7	104	285	438	(9)
	CDD50	4137	25	61	154	188	355	537	744	745	675	412	178	63	(10)
	CDD65	1014	0	0	2	4	37	123	282	282	232	51	1	0	(11)
	CDH74	10221	0	0	8	24	376	1297	3246	2903	2076	290	1	0	(12)
	CDH80	4395	0	0	0	0	90	499	1598	1322	844	42	0	0	(13)
(14) Wind	WSAvg	9.7	9.1	9.7	10.6	10.5	9.9	9.8	9.9	10.2	9.1	8.9	9.2	9.1	(14)
(15) Precipitation	PrecAvg	25.1	3.5	3.1	2.1	2.2	1.9	1.0	0.3	0.2	1.1	2.8	3.2	3.5	(15)
	PrecMax	40.9	13.5	8.2	8.2	6.9	4.8	4.1	2.9	2.9	3.0	8.5	9.6	9.8	(16)
	PrecMin	14.4	0.0	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.2	(17)
	PrecStd	7.0	3.1	2.5	1.9	1.5	1.4	1.1	0.6	0.5	0.9	2.4	2.6	2.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.5	66.6	74.5	77.2	87.2	93.9	100.2	99.6	97.8	84.2	71.3	63.8	(19)
		MCWB	52.4	55.3	58.0	58.4	62.8	65.2	69.1	69.2	68.5	62.4	58.7	54.1	(20)
	2%	DB	57.5	62.6	70.6	73.3	82.3	89.8	96.6	95.4	92.9	80.5	67.5	60.5	(21)
		MCWB	51.7	53.7	56.4	57.4	60.7	65.9	68.4	67.8	65.9	62.5	58.5	55.6	(22)
	5%	DB	55.8	59.4	67.0	69.8	78.5	85.9	93.0	90.9	88.7	77.3	64.8	58.6	(23)
		MCWB	51.4	52.3	54.9	55.9	59.9	64.3	67.3	66.1	65.0	60.7	57.9	54.7	(24)
	10%	DB	54.5	57.3	64.0	66.6	73.8	81.7	89.2	87.3	84.8	73.5	62.3	56.7	(25)
		MCWB	50.5	51.8	53.7	54.9	58.4	63.2	66.6	65.2	64.2	59.8	57.1	53.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.7	58.1	60.0	61.1	64.1	68.9	71.8	70.7	69.3	65.7	64.7	60.1	(27)
		MCDB	57.2	62.9	68.7	71.1	81.3	87.4	91.8	93.0	87.7	78.2	65.9	60.8	(28)
	2%	WB	54.4	56.1	57.7	59.3	62.1	66.9	69.9	69.0	68.0	64.1	60.9	57.9	(29)
		MCDB	55.8	59.3	67.0	68.6	77.7	85.2	91.2	90.9	83.9	74.4	63.8	59.0	(30)
	5%	WB	52.8	54.6	56.1	57.6	60.8	65.5	68.6	67.8	66.7	62.8	59.6	55.7	(31)
		MCDB	54.5	57.6	64.9	65.8	75.0	82.7	89.2	87.3	81.7	71.2	62.6	57.4	(32)
	10%	WB	51.2	53.0	54.6	56.3	59.4	64.0	67.2	66.5	65.7	61.6	58.3	53.9	(33)
		MCDB	53.5	56.0	62.1	63.8	71.1	79.1	86.6	83.1	79.9	69.4	61.4	55.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.9	14.3	14.4	17.1	20.2	23.4	22.7	19.2	14.3	10.5	9.4	(35)
		MCDBR	12.0	13.9	18.7	19.7	24.2	27.3	26.6	26.8	23.6	19.7	13.2	11.1	(36)
		MCWBR	7.9	7.8	8.9	8.8	8.9	9.5	8.1	8.6	7.8	7.3	6.5	7.2	(37)
	5% WB	MCDBR	9.4	11.3	16.4	16.7	21.4	24.6	25.8	24.9	19.9	15.6	9.7	7.6	(38)
		MCWBR	7.2	7.5	8.6	8.2	8.7	9.2	8.6	8.8	7.6	6.9	5.7	6.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.309	0.324	0.354	0.363	0.371	0.374	0.355	0.371	0.367	0.345	0.321	0.309	(40)
	taud		2.510	2.466	2.390	2.371	2.377	2.383	2.413	2.376	2.408	2.499	2.523	2.518	(41)
	Ebn at Noon		277	286	287	290	288	286	291	284	279	275	270	268	(42)
	Edh at Noon		24	29	35	38	38	38	36	37	34	28	24	23	(43)
(44) All-Sky Solar Radiation	RadAvg		733	1035	1387	1746	2108	2390	2480	2199	1695	1163	812	654	(44)
	RadStd		97	128	155	166	186	133	78	76	109	103	102	76	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page